National Research University Higher School of Economics (HSE)

Basic Curriculum

Field of study 01.04.02 Applied Mathematics and Informatics
Educational Programme "Math of Machine Learning"
Trajectories: "Math of Machine Learning"
Implementing unit: Faculty of Computer Science, HSE - Moscow

APPROVED
10.04.2025
Vice Rector
ROSHCHIN S.Y.
Signed with EDS

Years of Study: 2025/2026 - 2026/2027
Length of Programme: 2 years
Mode of Study: Full Time
Degree: Master's degree / MBA

| Block Code | Course | Subject type | Credits | Credits by Years | | Planned Educational Programme Development Results |
|---|---|---|---|---|---|---|
| | | | | 1 | 2 | |
| | Degree Programme | | 120,00 | 60,00 | 60,00 | |
| | Math of Machine Learning (General track) | | 120,00 | 60,00 | 60,00 | |
| | Key Seminars | | 15,00 | 6,00 | 9,00 | |
| 1 | Mentor's Seminar "Math of Machine Learning " (offered in a foreign language) | C | 15,00 | 6,00 | 9,00 | GPC-3.AMI, GPC-4.AMI, PC-1, PC-2, UC-2, UC-3, UC-4 |
| | Internship | | 30,00 | 6,00 | 24,00 | |
| | Preparation of the Graduate Thesis | | 24,00 | | 24,00 | |
| 1 | Graduation Thesis (offered in a foreign language) | C | 24,00 | | 24,00 | GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, GPC-4.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6 |
| | Fixed | | 6,00 | 6,00 | | |
| 1 | Innovation Workshop (offered in a foreign language) | C | 6,00 | 6,00 | | GPC-2.AMI, GPC-4.AMI, PC-1, PC-3, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6 |
| | Major | | 69,00 | 48,00 | 21,00 | |
| | Basic Components | | 30,00 | 27,00 | 3,00 | |
| 1 | Additional Topics in Probability Theory (offered in a foreign language) | C | 3,00 | 3,00 | | GPC-1.AMI, PC-6, UC-6 |
| 2 | Markov Chains (offered in a foreign language) | C | 3,00 | 3,00 | | GPC-1.AMI, PC-1, PC-2, UC-6 |
| 3 | High Dimensional Probability and Statistics (offered in a foreign language) | C | 6,00 | 6,00 | | GPC-2.AMI, PC-3, UC-6 |
| 4 | Machine Learning (offered in a foreign language) | C | 6,00 | 6,00 | | GPC-2.AMI, GPC-3.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, UC-6 |

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| 5 | Computational Linear Algebra (offered in a foreign language) | C | 6,00 | 6,00 | | GPC-1.AMI, PC-3, UC-6 |
| 6 | Sampling and Generative Modeling (offered in a foreign language) | C | 3,00 | 3,00 | | GPC-2.AMI, GPC-3.AMI, PC-1, PC-3, UC-1, UC-2, UC-3, UC-4 |
| 7 | Modern Algorithmical Optimization (offered in a foreign language) | C | 3,00 | | 3,00 | GPC-2.AMI, PC-1, PC-4, UC-1 |
| | Elective Courses. 1 year | | 21,00 | 21,00 | | |
| 1 | Introduction to Data Analysis (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-3.AMI, PC-4, UC-2 |
| 2 | Computational Imaging (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-2.AMI, PC-1, PC-4, PC-6, UC-2 |
| 3 | Foundations of Software Engineering (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-4.AMI, PC-5, UC-2 |
| 4 | Principles of Applied Statistics (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-1.AMI, PC-3, UC-1 |
| 5 | Introduction to Computer Vision (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-2.AMI, PC-2, PC-4, UC-3 |
| 6 | Optimization in Machine Learning (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-2.AMI, PC-1, UC-1 |
| 7 | Perception in Robotics (offered in a foreign language) | E | 6,00 | 6,00 | | GPC-2.AMI, PC-3, PC-4, UC-2 |
| 8 | Tensor Decompositions and Tensor Networks in Artificial Intelligence (offered in a foreign language) | E | 6,00 | 6,00 | | GPC-2.AMI, PC-1, PC-3, UC-1 |
| 9 | Deep Learning (offered in a foreign language) | E | 6,00 | 6,00 | | GPC-4.AMI, PC-1, PC-4, UC-2, UC-6 |
| 10 | Geometrical Methods of Machine Learning (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-2.AMI, PC-1, UC-1 |
| 11 | Deep Learning for Natural Language Processing (offered in a foreign language) | E | 3,00 | 3,00 | | GPC-1.AMI, PC-1, PC-2, PC-3, UC-2 |
| | Elective Courses. 2 year | | 18,00 | | 18,00 | |
| 1 | Mathematical Foundations of Reinforcement learning (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-1.AMI, GPC-2.AMI, PC-3, PC-6, UC-6 |
| 2 | Optimal Transport and Its Applications (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, PC-1, PC-3, PC-4, UC-1 |
| 3 | Leadership for Innovators (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-4.AMI, PC-6, UC-1, UC-3, UC-4, UC-6 |
| 4 | Generative Models in Machine Learning (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-4.AMI, PC-5, PC-6, UC-2, UC-6 |
| 5 | Reinforcement Learning (offered in a foreign language) | E | 6,00 | | 6,00 | GPC-2.AMI, PC-1, PC-3, UC-2 |
| 6 | High Performance Python Lab (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-3.AMI, PC-3, PC-5, UC-1 |
| 7 | Planning Algorithms in Artificial Intelligence (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-2.AMI, PC-1, PC-3, PC-6, UC-1 |

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| 8 | Theoretical Methods of Deep Learning (offered in a foreign language) | E | 3,00 | | 3,00 | GPC-1.AMI, GPC-2.AMI, UC-1, UC-2 |
| | Final State Certification (FSC) | | 6,00 | | 6,00 | |
| 1 | Final Qualification Paper (offered in a foreign language) | C | 6,00 | | 6,00 | GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, GPC-4.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6 |

***Curriculum agreed:***

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| Academic Supervisor | Naumov A.A. | 04.04.2025 |
| Dean | ARZHANTSEV I.V. | 07.04.2025 |
| Head of Centre for Educational Model Design | LEPESHKIN I.A. | 09.04.2025 |

Образовательная программа направлена на формирование и развитие следующих компетенций (результатов освоения):

| Competence code | Name of the competence |
|---|---|
| UC-1 | Способен осуществлять критический анализ проблемных ситуаций на основе системного подхода, вырабатывать стратегию действий. |
| UC-2 | Способен управлять проектом на всех этапах его жизненного цикла. |
| UC-3 | Способен организовывать и руководить работой команды, вырабатывать командную стратегию для достижения поставленной цели. |
| UC-4 | Способен применять современные коммуникативные технологии, в том числе на иностранном(ых) языке(ах), для академического и профессионального взаимодействия. |
| UC-5 | Способен анализировать и учитывать разнообразие культур в процессе межкультурного взаимодействия. |
| UC-6 | Способен определять и реализовывать приоритеты собственной деятельности и способы ее совершенствования на основе самооценки и образования в течении всей жизни. |
| GPC-1 | Способен решать актуальные задачи фундаментальной и прикладной математики. |
| GPC-2 | Способен совершенствовать и реализовывать новые математические методы решения прикладных задач. |

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| GPC-3 | Способен разрабатывать математические модели и проводить их анализ при решении задач в области профессиональной деятельности. |
| GPC-4 | Способен комбинировать и адаптировать существующие информационнокоммуникационные технологии для решения задач в области профессиональной деятельности с учетом требований информационной безопасности. |
| PC-1 | Able to find relevant areas of scientific research, set up and test scientific hypotheses |
| PC-2 | Able to analyze product tasks and propose solutions based on data |
| PC-3 | Able to build hypotheses and test them based on data |
| PC-4 | Able to choose according to the task being solved and build machine learning models, as well as evaluate their quality |
| PC-5 | Able to understand the architecture of large projects and implement new functionality in them |
| PC-6 | Able to apply in practice the acquired fundamental and applied knowledge for the selection of an actual task and the organization of the development process |

* Subject type:

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| Compulsory course | C |
| Elective course | E |